



Mathieu ASTRUC

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Available from Oct. 2026

EXPERIENCE

Airbus

Mar. 2026 – Sep. 2026

AI Engineer - Intern – Master Thesis (ongoing)

France

- Designing and building a RAG-powered, domain-specific AI chatbot for the Export Controls function in a safety-critical environment.
- End-to-end pipeline: covering ingestion, embedding/retrieval, orchestration, evaluation, token-cost reduction, user feedback integration, and deployment preparation with the lead architect.
- Developing Proofs of Concept for AI-driven solutions in collaboration with cross-functional teams, supporting the exploration and integration of enterprise AI use cases.

NTNU – Norwegian University of Science and Technology

May 2025 – Aug. 2025

Research Engineer - Intern

Norway

- Developed an embedded AI interaction stack for a humanoid robot, combining real-time gesture recognition, computer vision and a fine-tuned LLM for domain-specific dialogue.
- Optimized on-device inference latency through GPU pipeline tuning, enabling real-time responses under hardware and runtime constraints.
- Implemented a Human-in-the-Loop framework to manage edge cases, improve robustness and support continuous model improvement.

Banque de France

May 2024 – Aug. 2024

Data Scientist - Data Engineer - Intern

France

- Automated scraping workflows aggregating unstructured public/financial data from multiple sources.
- Combined OCR, Speech-to-Text and LLMs to digitize raw audio/visual content, feeding a downstream RAG system.

Comat Specific

Jan. 2025 – May 2025

Machine Learning Engineer – Part-time role alongside studies

France

- Built a deep-learning OCR pipeline converting legacy hand-drawn 2D engineering sketches into structured machine-readable data, modernizing industrial workflows.

SELECTED PROJECTS

Publication – HCI International 2026, Montreal (Springer Conference Proceedings)

2025 – 2026

- Lead author of an accepted paper on real-time hand gesture recognition for human-robot interaction.

Founder – Digital Content Project (200k+ followers, 100M+ views across platforms)

2024 – Present

- Built and scaled content brands from scratch across multiple niches on YouTube, Instagram, TikTok and Facebook, now focused on AI, Data Science and Tech (30k+ followers) – full ownership of editorial line, scripting, production and growth, sharpening the ability to translate complex technical topics for broad audiences.

EDUCATION

Esaip – Master of Engineering, Data Science (Grade A - Top 10%)

Sep. 2021 – Sep. 2026

Engineer's Degree (5-year program) - Optimization algorithms, Multicriteria Optimization, ML, Image Processing, Reinforcement Learning, Explainable AI, Probabilistic Modeling...

France

Universidad Politécnica de Madrid

Sep. 2025 – Feb. 2026

Study Abroad – Machine Learning, cloud computing, large-scale data architectures

Spain

SeAMK – University of Applied Sciences

Jan. 2024 – Jun. 2024

Study Abroad – Machine Learning, Deep Learning, Software Development, Embedded Systems

Finland

SKILLS

Programming Languages Python, C, C++, SQL, Bash

Data Science & Machine Learning PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Transformers

LLM engineering LangChain, LlamaIndex, Agent workflows, Fine-tuning (LoRA / QLoRA), Vector DBs (FAISS, Chroma)

Specializations NLP, RAG, Computer Vision, Multimodal AI, Human Robot Interaction

Inference & Deployment vLLM, llama.cpp, Hugging Face TGI, FastAPI, Docker, Kubernetes

Languages French (native), English (C1), Spanish (B1)

BEYOND WORK

Former competitive basketball player · Entrepreneurship · Chess · Sports · Research